

THE FUTURE OF COMPUTING

Innovations for the Future of Computing

Moore's Law predicts that the number of transistors in an integrated circuit doubles every two years as technological advances continue. It means that every two years, electronic devices can do twice as many new and innovative functions as before.

It was observed by Gordon Moore in 1965, who saw that the size of transistors was rapidly shrinking due to continuous innovation.

For example, a computer chip that contained 2,000 transistors that cost P50,000 in 1970 became P25,000 in 1972, P12,500 in 1974, and P48.50 in 1990. Nowadays, it costs less than P1.00 to manufacture today.

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However, in 2015, Gordon Moore stated that Moore's Law would not be as reliable in the next decade as companies slowly shift focus toward non-silicon computing which may result in substantial efficiency gains soon.

These are the following innovations in the computing industry:

1. Graphene-based Transistors

Graphene is a thin one-atom-thick sheet that is more conductive than any other known material. It can be rolled up into tiny tubes called *nanotubes* and combined with other 2D materials to move electrons quickly, requiring lesser space and energy than the smallest silicon transistor.

Producing nanotubes is too complicated and error-prone to be commercially viable. But in 2019, a process for creating a 16-bit carbon nanotube microprocessor was developed. It successfully executed a set of commands that printed the message, "*Hello, World!*" It removed limitations and complications in nanotubes, making them accessible to factories in less than five (5) years.

2. Quantum Computing

It uses quantum bits, or qubits, which are the physical carrier of quantum information. Compared to a traditional bit, one (1) qubit is equivalent to two (2) bits meaning that 2^n where n is the qubit value.

Using subatomic particles, quantum computers can represent different states simultaneously. These particles will then be controlled by the rules of probability and not absolute states or logic gates.

Controlling small subatomic particles allow researchers to solve problems, such as identifying drug properties, performing complex computations, and helping chip designers create impossibly complex circuits.

3. DNA Computing

As computers can only analyze one potential answer at a time, the advancements in DNA computing due to parallel processing made analyzing all possible answers at once available. It is possible as DNA data storage can put all the information into the tiniest storage.

The developments are extraordinarily stable as scientists used a scrap of bone to reconstruct the genome of a cave bear that died 300 millennia ago. Other features include DNA fingerprinting and decoding strategic information such as banking, military, and communications data.

4. Neuromorphic Technology

It aims to create a computer that imitates the architecture of the human brain with much lesser energy than a traditional transistor.

In 2020, Intel released a new server based on neuromorphic chips with the same neural capacity as a small mammal's brain. International researchers have already linked artificial and biological neurons to communicate like a biological nervous system based on internet protocols.

5. Optical Computing

It uses photons by mapping data onto light-intensity levels and then changing the light intensity to perform calculations allowing optical computing at the nanoscale possible at the speed of light. Although still in an early development phase, this could enable high-efficiency and low-power processing data transmission

Optical computers may use fiber-optic cables, optical chips, or wireless optical networks to process and transmit data.

6. Distributed Computing

The current peak of this model is the *Folding@home* which models protein molecules to discover cures for serious diseases such as Alzheimer's, cancer, and COVID-19.

The project has amassed nearly 750,000 participants with a collective of 1.5 exaflops of power, allowing quintillion calculations per second. It has 75% of the projected speed of the *El Capitan supercomputer*, which will be deemed the world's fastest supercomputer by 2023.

The future of computing will be shaped by transistors made from materials other than silicon amplified by approaches unrelated to transistor speed, such as deep-learning software and the ability to crowdsource excess computing power to create supercomputers.

All the technology and innovations for the future mentioned may even redefine computing itself as we know it.

Collapse OS: Operating System After the World Ends

A programmer has decided that creating a versatile and survivable operating system is as important as stocking food and other living essentials after a societal collapse. This operating system is called **Collapse OS**.

It is an open-source operating system made explicitly once the worse has happened on Earth. According to Virgil Dupras, a software developer, people will need Collapse OS to reconfigure their damaged smartphones.

Creation

In 2019, Dupras envisioned the collapse of the global supply chain by 2030, where populations will not be able to produce electronics anymore. He predicts that political and social power remains and that anyone who can scavenge electronic devices will have huge advantages.

Features and Functions

Collapse OS will run on Z80 8-bit microprocessors. Z80 8-bit are found in desktop computers, cash registers, and graphing calculators, though less common today than 16- and 32-bit components.

Collapse OS will allow the following:

- Run on minimal and improvised machines.
- Interface through improvised means like a keyboard and display.
- Compile assembler source files
- Read and write from a range of storage devices.
- Edit text files.
- Replicate itself.

It can self-replicate with enough RAM and storage, read SD cards, and has a simple text editor. The kernels are modules connected with glue code to make the system flexible and adaptable.

Here is the detailed roadmap for the direction of the project:

- Support for 8080 and 6502 CPUs
- Support LCD screens and E-ink displays.
- Support storage options such as floppies, CDs, and RAM/ROMs.
- Support for TI-83+ and TI-84+ graphing calculators and TRS-80s.

Conclusion

The goal is to initiate a post-collapse civilization's return to the computer age.

Electronics production relies on a complex supply chain that, once it crumbles, will return humankind to a less technical age and face difficulties in regaining the previous technical position. Dupras hopes to skip several steps by creating an operating system that will work with simpler chips that can be scavenged from various sources.

References:

- CFI (2020). *Moore's law*. [Web Article]. Retrieved on July 25, 2022, from <https://corporatefinanceinstitute.com/resources/knowledge/other/moores-law/>
- CFI (2022). *Computers, Future of*. [Web Article]. Retrieved on July 25, 2022, from <https://www.encyclopedia.com/education/news-wires-white-papers-and-books/computers-future>
- Paul, J. (2019). *Collapse OS: An OS created to run after the world ends* [Web Article]. Retrieved on July 25, 2022, from <https://itsfoss.com/collapse-os/>
- Wellers, D. (n.d.a.). 6 surprising innovations for the future of computing. [Web Article]. Retrieved on July 25, 2022, from <https://insights.sap.com/6-surprising-innovations-for-the-future-of-computing/>